A Survey of Constraints for the Examination Timetabling Problem

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Abstract

Exam timetabling in higher education involves numerous regulatory and organizational constraints that are often institution-specific. Despite many algorithmic approaches showing promising results on simplified datasets, practical adoption of automatic scheduling tools remains limited due to a lack of alignment with real-world needs. This paper presents a comprehensive catalogue of constraints collected from multiple institutions. The aim is to improve modularity in scheduling models and to facilitate communication between IT teams and stakeholders, thereby easing the adoption of automated scheduling solutions.

Keywords: Exam timetabling, Scheduling, Constraint programming, Higher education

1 Introduction

Examinations are a central component of academic programs. They generate significant stress for students (Ogden and Mtandabari (1997)) and place considerable pressure on institutional resources. To mitigate these challenges and comply with stringent regulations, institutions must allocate substantial human resources to construct examination timetables that satisfy regulatory requirements, human resource constraints, and student well-being considerations, all within limited material capacities.

With the advent of computers in administration, IT departments and research teams have sought to automate part, if not all, of the examination scheduling process (Burke et al. (1996)). While numerous algorithms have demonstrated satisfactory performance within theoretical frameworks and on simplified datasets (Qu et al. (2006)), practical adoption within institutions remains limited. This gap is partly due to the misalignment between existing software solutions and the heterogeneous, institution-specific constraints encountered in real-world examination scheduling (McCollum (2007)).

The Université de Tours (France) was considering the development and adoption of an automated system for exam scheduling, with the goal of reducing the workload on faculty and administrative staff and allowing them to focus on less tedious tasks in order to improve work conditions and service quality. As part of this effort, the university commissioned a feasibility study, which included the creation of a prototype. To ensure the prototype's relevance, it was necessary to compile a list of constraints currently encountered or potentially encountered by the institution. This initial internal list[1], compiled by Andreas Mulard in 2025 as part of the feasibility study, formed a smaller, unindexed version of the catalogue presented in this article.

In this paper, we address this issue of constraint understanding by proposing a diverse, comprehensive, sourced, and indexed catalogue of examination scheduling constraints, together with a bibliography that supports traceability of each constraint to its institutional origin. This catalogue is designed to promote modular modeling approaches and to facilitate communication between IT teams and the various stakeholders involved in examination scheduling. The indexing provided reduces the risk of miscommunication across teams and documents.

2 Related works

Existing constraint collections have typically been conducted at the departmental or institutional level, such as Cowling et al. (2002) for MARA University of Technology. The most ambitious constraint collection effort was conducted by Burke et al. (1996), who gathered information on constraints and practices related to examination scheduling through a survey of ninety-five British universities. More recently, Siew et al. (2024) surveyed methodologies for the exam timetabling problem and published a small set of constraints.

Nonetheless, most sources of constraints in the exam scheduling problem come from articles focused on solution methods, where researchers use constraints from their own institutions to build plausible models for evaluating the effectiveness of their approaches, such as Bergmann and Zurheide (2014).

A recent case study by Chaplin et al. (2025) likewise reports that mismatches between institutional practices and off-the-shelf scheduling tools remain a primary barrier to adoption.

3 Methodology

3.1 Constraint Collection

For this survey, we developed three approaches to constraint collection. The first consists of interviews with the administrative teams at the Université de Tours. The second is based on the analysis of models used in scientific articles. Finally, we constructed additional constraints by deriving variations from existing ones.

[1] Available at https://andreasmulard.fr/recherche/catalogue_contrainte.pdf

3.1.1 Survey at Université de Tours

The first source of constraints comes from fieldwork conducted at the Université de Tours, with a focus on the Faculty of Science and Technology and the central Teaching Directorate (DIFOR). The objective was to collect real, institution-specific constraints encountered in the exam scheduling process.

Interviews were conducted with administrative staff at the Faculty of Science and Technology. These discussions revealed the practical constraints that arise in scheduling examinations across a variety of programmes, including special diplomas and students with specific statuses (e.g., double degrees, part-time students). The resulting list of constraints was reviewed and validated by faculty administrators to ensure it accurately reflects operational requirements.

During the course of the research, discussions were held with staff members from DIFOR, the university's central teaching directorate. Among its many responsibilities, DIFOR provides support to faculty administrative teams in managing the timetabling software. These interviews uncovered additional constraints not present in the Faculty of Science but commonly encountered in other departments. The inclusion of these broader constraints helped to generalize the catalogue beyond the initial faculty context and ensured wider applicability across institutional settings.

3.1.2 Extraction from Research Literature

The second source of constraints was the academic literature on examination timetabling and related scheduling problems. This literature is scattered across several fields, including computer science, operations research, mathematics, sociology, and management. As a result, no single journal, conference, or bibliographic database provides comprehensive coverage of the constraints used in practice or reported in models. In this section, a paper is said to have been reviewed when its title, abstract, keywords, and, when necessary and accessible, full text were examined in order to identify explicitly stated or implicitly required examination timetabling constraints.

Google Scholar search.

One source consisted of a Google Scholar search using the terms "exam timetabling" and "exam scheduling". This search was not restricted to a specific publication period. Papers were reviewed until saturation was reached on July 7, 2025, with saturation defined as four consecutive reviewed papers introducing no new constraints. In most articles, constraints were introduced only as contextual elements used to evaluate algorithmic performance; nevertheless, any explicitly stated or implicitly required constraint was extracted and incorporated into the catalogue.

Keyword-based scan of conferences and journals.

A second source consisted of a keyword-based scan of major outlets in automated timetabling, scheduling, and planning. The scan was conducted on Nov. 15, 2025. For PATAT, MISTA, and the Journal of Scheduling, we examined all accessible papers or articles available up to that date that contained the keyword "exam" in the searchable metadata or text. For the journals listed in Appendix A, publisher portals were queried using examination-related keywords, and all accessible relevant articles available up to Nov. 15, 2025 were considered. We also reviewed examination-related scheduling material from CPAIOR, from ICAPS proceedings between the 13th and 34th editions, from the Conference on Artificial Intelligence Planning Systems excluding the first edition, from the 4th and 5th editions of the European Conference on Planning, and from the 1990 Expert Planning Systems workshop.

Contribution to the catalogue.

These two complementary literature-based sources enabled the identification of constraints originating from diverse educational systems and modelling traditions. They enriched the catalogue beyond the local institutional context and improved its general applicability.

3.1.3 Constructed Constraints

To enhance the completeness and usability of the constraint catalogue, we introduced a set of trivial constructed constraints derived from simple transformations or generalizations of existing ones. These constraints were not explicitly identified in either the institutional survey or the literature review, but they logically follow from common patterns and scheduling logic.

For instance, the constraint (D1-1) “Exam A shall be scheduled before Exam B”, observed at Université Catholique de l’Ouest (Boizumault et al. (1996)), was complemented by its inverse (D1-2) “Exam A shall be scheduled after Exam B”, which, while not directly reported, is a trivial and practically relevant counterpart.

Constraints constructed in this way are explicitly labeled as “constructed constraints” in the catalogue, to distinguish them from those directly sourced from institutional or academic references.

3.2 Constraint Categorization and indexing

In many studies, such as Ayob et al. (2007), constraints are classified into two main types: hard constraints, whose violation renders a timetable infeasible, and soft constraints, whose violation only reduces the quality or objective score of a solution. However, this classification is not universal. Its interpretation depends heavily on the institutional context. For example, some universities may forbid examination merging (R4-1), while others may allow it with a penalty (R4-2). At a minimum, each institution must treat its legal and regulatory obligations as hard constraints, yet such obligations vary considerably across contexts.

Accordingly, our catalogue is organized using a hierarchical classification system:

- At the top level, constraints are grouped by broad functional domains, such as room-related constraints, denoted by the code (R).

- At the second level, each domain is subdivided into specific subcategories, each capturing a particular set of scheduling rules, such as (R2) “Equipment / Environment”.

Within each subcategory, we list variants of constraints drawn from different institutions and publications, in order to reflect the diversity of how similar issues are handled across contexts, such as a room must have a minimum number of equipment (R2-5).

To support this structure, each constraint is assigned an identifier following the pattern Kx–y, where the letter denotes the domain, the first index x refers to the subcategory, and the second index y identifies the specific constraint variant.

Some constraints may be relevant to more than one category. In such cases, they are listed under multiple headings and explicitly marked as equivalent constraints to indicate their conceptual similarity.

There is no universally accepted standard for expressing constraints, and it is often difficult, if not impossible to formulate them in a fully neutral way that does not implicitly suggest whether they should be treated as hard or soft constraints. Consequently, some formulations in this catalogue may reflect these implicit distinctions, which we accept as a practical limitation. This is not considered problematic, as implementing teams can readily adapt or reformulate constraints to suit their specific modelling choices.

Below is an overview of the main top-level categories used in the catalogue:

3.2.1 (R) Room and Facility

This category includes constraints related to the physical characteristics of examination venues: room capacity, required equipment, environmental conditions (e.g., lighting or accessibility), spatial preferences, and room-sharing limitations.

3.2.2 (S) Student Conflict and Spread

This category covers both conflict-avoidance constraints (e.g., a student cannot be scheduled for overlapping exams) and spread constraints that aim to reduce student workload over short time windows (e.g., "no more than one exam per day" or "minimum two-hour gap between exams"). They promote equity and student well-being, which are critical for timetable acceptance.

3.2.3 (I) Invigilation and Staffing

These constraints govern the human resources involved in exam supervision. They include minimum staffing levels per room, invigilator availability, required skills (e.g., language proficiency), rest periods, and fair workload distribution. Integrating these constraints early in the scheduling process helps avoid conflicts with institutional HR policies.

3.2.4 (D) Sequencing and Dependencies

Sequencing constraints define the required order between exams whether based on pedagogical logic (e.g., prerequisites) or structural dependencies (e.g., joint assessment of lab and theory components). These constraints preserve curricular coherence and prevent logical inconsistencies in exam ordering.

3.2.5 (E) Equity and Inclusion

This category includes constraints ensuring accommodations for students with disabilities, religious observances, or linguistic requirements. Encoding these needs explicitly not only ensures compliance with legal frameworks but also reinforces institutional commitments to inclusion and diversity.

3.2.6 (L) Logistics and Cost

Finally, this category groups constraints aimed at improving operational efficiency. Examples include avoiding exam sessions during public transport peak hours, minimizing building use after hours, or reducing the use of costly resources. These are typically modeled as soft constraints and provide decision-makers with quantitative levers to balance cost, comfort, and accessibility.

4 Results

Our survey resulted in a set of **167 individual constraints**, distributed across six main categories.

Table 1 Distribution of constraints by category

Category	Number of Constraints
Room & Facility	28
Student Conflict & Spread	27
Invigilation & Staffing	38
Sequencing & Dependencies	26
Equity & Inclusion	18
Logistics & Cost	30
Total	**167**

The constraints were collected from three types of sources: extraction from research literature, fieldwork at the Université de Tours, and constructed constraints derived from existing ones.

Table 2 Distribution by source of constraint

Source of Constraint	Number of Constraints
Extraction from Research Literature	132
Survey at Université de Tours	32
Constructed Constraints	3
Total	**167**

Some constraints were identified through multiple collection methods. In such cases, each constraint was counted only once and assigned to the strongest source category according to the following priority order: survey at Université de Tours, extraction from research literature, and constructed constraints. This hierarchy reflects the level of evidence associated with each source. The institutional survey provides evidence of current or recent real-world use, the literature review documents reported use cases in academic studies, and constructed constraints represent plausible potential use cases derived from existing ones.

Europe remains by far the best-represented region in our catalogue, followed by North America and the Middle East / Gulf. Asia-Pacific accounts for a modest share of cases, while South America and Africa are present only through single institutions (Chile and Tunisia, respectively). No institutions from Oceania appear in the current dataset. Several large higher-education systems, such as those of Japan, Brazil, India, Pakistan, Mexico, Germany, and Russia, are still missing, pointing to clear priorities for future data collection and catalogue expansion.

Table 3 Distribution of sources by country

Country / Region	Sources
France	4
United Kingdom*	1
Malaysia	4
Canada	4
Turkey	3
United States	4
Czech Republic	1
Saudi Arabia	2
Kuwait	1
Chile	1
Belgium	1
Greece	1
Hungary	1
Norway	N/A
United Arab Emirates	2
Tunisia	1
China	1
Ireland	1
Israel	1

* The group of UK institutions reported in Burke et al. (1996) is counted as one.

The sample spans institutions of varying sizes and governance models, including public, private, faith-based, and military institutions, thereby providing a diverse foundation for future benchmarking efforts.

Some constraints listed in the catalogue may no longer be applicable, while others may apply only to specific faculties within an institution. When a source indicated a constraint already present in the catalogue, it was not duplicated in order to limit the size of the appendices.

Table 4 Distribution of sources by geographic region

Region	Sources
Africa	1
Asia–Pacific	5
Europe*	12
Middle East & Gulf	9
North America	8
South America	1

* The group of UK institutions reported in Burke et al. (1996) is counted as one.

Table 5 Distribution of sources by institutional status

Status Category	Sources
Public	26
Private	6
Confessional / Faith-based	2
Military	1

Table 6 Distribution of sources by student enrollment

Size Band	Sources
$<$10 000	8
10 000–29 999	16
$\geq$30 000	10

5 Constraint catalogue

The complete catalogue of constraints is presented below. For each constraint, the code indicates its category, subcategory, and variant. The origin of the constraints is reported in Appendix B[2].

Table 7: Constraints in the "Room & Facility" category

Code	Description
R1 Capacity	
R1-1	The seating capacity of the assigned room shall not be lower than the number of registered candidates.
R1-2	The student-to-seat ratio for any examination shall not exceed a pre-defined limit.
R1-3	The total number of candidates examined on a given calendar day shall not exceed an institution-specific ceiling.
R1-4	The aggregate number of students present within a single building at any time shall not surpass a capacity limit set by safety regulations.

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[2]Also available at https://github.com/Kaputchino/Constraints-for-the-Examination-Timetabling-Problem.

Code	Description
R1-5	For oral examinations, the number of examinees in a time slot shall not exceed the maximum interview capacity authorised by the department.
R1-6	The number of distinct examinations assigned to the same room and time slot shall not exceed a predefined maximum N.
R1-7	No more than N students may sit an exam at any one time.
R2 Equipment / Environment	
R2-1	A buffer of not less than a specified time interval shall separate consecutive examinations scheduled in the same room.
R2-2	The temperature shall remain within an acceptable range, or the room must offer appropriate climate control.
R2-3	The room shall remain continuously available for the entire duration of the examination.
R2-4	The room must have an equipment.
R2-5	The room must have an equipment for at least a predefined share of students taking the exam.
R3 Location / Preference	
R3-1	Preferentially assign examinations to rooms located within the candidates' departmental building.
R3-2	Certain examinations shall be scheduled exclusively in designated rooms.
R3-3	Large exam halls must be scheduled in preference to smaller ones.
R3-4	Exams must be scheduled to rooms near the relevant department.
R3-5	Exams for the same department should be kept together.
R3-6	Specific rooms may be reserved exclusively for particular degree programmes.
R3-7	A student who has consecutive exams on the same day should be assigned to the same room.
R3-8	A penalty is incurred when examinations for distinct courses are placed in the same building or in adjacent rooms, so as to encourage spatial diversity.
R3-9	Rooms allocated to a split examination shall, where possible, be located in the same building or campus sector.
R4 Merger	
R4-1	The examination shall not be merged with any other examination.
R4-2	The examination may be merged with other examinations.
R4-3	Only examinations belonging to authorised categories may be merged.
R4-4	Oral and written components shall not be merged into the same room.
R4-5	Candidates shall be split into separate rooms according to examination end times.
R4-6	Examinations of differing duration shall not be scheduled in the same room.
R4-7	The number of rooms assigned to a split examination shall be minimised.

Table 8: Constraints in the "Student Conflict & Spread" category

Code	Description
S1 Load	
S1-1	A minimum gap of Δ hours shall separate any two examinations taken by the same student on the same day.
S1-2	Subjects designated as difficult shall be placed toward the end of the examination session to mitigate early cognitive overload.
S1-3	Each student's examinations should be evenly distributed across the timetable.
S1-4	No student shall be required to sit more than a set number of examinations per day.
S1-5	For a given academic programme, the number of examinations per day shall not exceed the programme-specific daily limit.
S1-6	The cumulative examination duration assigned to any student over any two consecutive days shall not exceed the maximum total duration set by policy.
S1-7	Within a single academic department, the number of examinations scheduled on the same day shall not exceed the departmental daily limit.
S1-8	If a student has only one examination on a given day, that examination should be placed outside the evening time slots to reduce on-campus idle time.
S1-9	The number of consecutive examinations assigned to a student within the same day shall not exceed the permitted consecutive-exam limit.
S1-10	Take-away examinations should not be scheduled concurrently with in-person examinations.
S1-11	A student shall not be assigned an examination in the last time slot of one day followed by another in the first slot of the next day, so as to ensure sufficient overnight rest.

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Code	Description
S1-12	The average difficulty of a student's examinations shall be fairly distributed.
S1-13	A penalty shall be applied to examinations scheduled on weekends.
S1-14	All examinations belonging to the same academic year shall be scheduled in a single designated time band (morning, afternoon, or evening).
S1-15	A scheduling penalty shall be applied when a student is assigned examinations in both the period before lunch and after lunch on the same day.
S1-16	Examinations for courses in the same programme and term shall not be scheduled in consecutive time slots on the same day.
S2 Overlap	
S2-1	A student shall not be allocated to two examinations occurring in the same time slot.
S2-2	Re-sit candidates must be prevented from overlapping with another examination.
S2-3	Examination clashes for each academic programme should be minimised.
S2-4	Examination clashes for each programme–year combination should be minimised.
S2-5	Scheduling should minimise clashes among elective courses.
S2-6	Exams A and B must not share the same period. (Equivalent to D1-4)
S2-7	Examinations for courses belonging to the same academic programme and term shall not share the same time slot.
S2-8	Minimise the disparity in exam-slot conflicts among academic programmes and terms.
S3 Assignment	
S3-1	Minimise the number of students assigned to examination commissions where neither their supervisor nor referee serves as examiner.
S3-2	Students supervised (or reviewed) by the same faculty member shall be scheduled in consecutive examination sequences so that the faculty member attends the minimum possible number of separate sequences. Equivalent to I1-10.
S3-3	The thesis should be scheduled such that both the supervisor and the referee can attend the examination. Equivalent to I3-6.

Table 9: Constraints in the "Invigilation & Staffing" category

Code	Description
I1 Workload	
I1-1	The cumulative invigilation workload assigned to any staff member shall not exceed the maximum load defined by institutional policy.
I1-2	Invigilation duties shall not be assigned in the late-evening time band more frequently than the frequency limit specified by the institution.
I1-3	Each instructor shall have at least one unscheduled day per examination week.
I1-4	Staff holding additional roles (e.g., administration or funded research) shall receive a reduced number of invigilation duties.
I1-5	The number of examinations invigilated by any staff member during the exam period shall not exceed the staff-duty limit specified by the institution.
I1-6	Invigilation duties shall be balanced among faculty members holding the same academic rank.
I1-7	Chief-invigilator duties shall be evenly spread among eligible lecturers.
I1-8	Invigilation assignments shall be distributed so that each invigilator carries a comparable total duty load.
I1-9	A minimum rest period of Δ hours shall separate consecutive invigilation duties assigned to the same staff member.
I1-10	Students supervised (or reviewed) by the same faculty member shall be scheduled in consecutive examination sequences so that the faculty member attends the minimum possible number of separate sequences. Equivalent to S3-2.
I1-11	Minimise the number of proctor schedules containing split shifts on the same day.
I1-12	Avoid evening-to-morning schedules.
I1-13	Distribute evening-to-morning schedules evenly among proctors.
I1-14	Allocate a greater number of invigilation shifts to proctors in higher experience classes.
I1-15	Within the same experience class, assign more shifts to proctors who are available for more shifts and willing to work in more rooms.
I1-16	Assign fewer unfavourable schedules to proctors in higher experience classes.
I2 Roles / Separation	
I2-1	An instructor shall not invigilate an examination for which they are the principal teacher.

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Code	Description
I2-2	Teaching staff shall not simultaneously hold the roles of teacher and invigilator for the same examination.
I2-3	Where pedagogical policy requires it, the same individual may serve as teacher, subject expert, supervisor, and invigilator for a designated examination.
I2-4	Of the professors assigned to a session, at least one shall be the principal examiner and at least one an external supervisor.
I2-5	There should be one instructor at each timeslot who can serve as chair.
I2-6	There should be one instructor at each timeslot who can serve as secretary.
I2-7	There should be at least one instructor at each timeslot who can serve as member.
I3 Availability	
I3-1	An invigilator shall be available for the entire duration of the examination to which they are assigned.
I3-2	A minimum buffer of Δ hours shall separate an instructor's own examination from any subsequent invigilation duty.
I3-3	Staff involved in project presentations or evaluations shall not be scheduled for invigilation during conflicting time blocks.
I3-4	A guard-time parameter Δ shall be observed before an invigilator may supervise a second examination.
I3-5	Instructor time-slot preferences should be honoured where feasible.
I3-6	The supervisor and the referee of a thesis examination must be scheduled so that both can attend. Equivalent to S3-3.
I3-7	Examinations for courses taught by the same instructional team shall be scheduled in different time slots.
I3-8	Car-pool members always work the same shifts.
I4 Requirement	
I4-1	Each examination shall be supervised by at least the minimum number of invigilators specified by regulation.
I4-2	Each academic department shall provide at least the departmental quota of proctoring personnel per examination day.
I4-3	Designated examinations shall be supervised by specifically named invigilators.
I4-4	Where required, female proctors shall be assigned to examinations held for female-only cohorts. (Equivalent to E2-2)
I4-5	Each campus shall supply invigilators for its own examinations using its assigned pool of teaching staff.
I4-6	Each shift shall include any full-time proctor who is available for that shift.
I4-7	A minimum of two professors qualified for the subject shall be assigned to every active examination session.

Table 10: Constraints in the "Sequencing & Dependencies" category

Code	Description
D1 Sequence	
D1-1	Exam A shall be scheduled before Exam B.
D1-2	Exam A shall be scheduled after Exam B.
D1-3	Exams A and B shall be scheduled in the same period.
D1-4	Exams A and B must not share the same period. (Equivalent to S2-6)
D1-5	Exams sharing common questions shall be scheduled concurrently.
D1-6	Exam A must precede Exam B by at least a pre-defined time interval Δ.
D1-7	Exam A must follow Exam B by at least a pre-defined time interval Δ.
D1-8	Exactly one exam in a designated group shall be scheduled last within that group.
D1-9	Exactly one exam in a designated group shall be scheduled first within that group.
D1-10	Specified pairs of exams shall be held on the same calendar day.
D1-11	Exams belonging to mutually exclusive categories (e.g., two music classes) shall not be scheduled simultaneously.
D1-12	Where feasible, avoid scheduling designated courses in specified examination periods.
D1-13	All written papers of the same subject shall be scheduled in a single period.
D1-14	Exams A and B shall be scheduled in the same room at the same time.
D2 Fixed Time	

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Code	Description
D2-1	An examination should be held at the same time as the regular lecture where institutional policy so requires.
D2-2	An examination must be scheduled no later than a pre-defined deadline.
D2-3	An examination must be scheduled no earlier than a pre-defined release time.
D2-4	An examination must be held at a fixed, pre-assigned period.
D2-5	The final examination for a course shall be scheduled only on weekdays that match its regular meeting pattern.
D3 Dependent Scheduling	
D3-1	Examinations with the largest candidate counts should be scheduled early in the timetable.
D3-2	High-capacity courses should be timetabled in the first week to maximise grading time for lecturers.
D3-3	Finals, honours examinations, or papers subject to external moderation should appear early in the session.
D3-4	Examinations with the largest candidate counts shall be scheduled toward the end of the examination session.
D3-5	Designated examinations shall rotate their position within the timetable across successive academic years so that no single exam is always last.
D3-6	Retain the optimised timetable from the previous year as much as possible by minimising changes.
D3-7	Examinations shall be fixed in hierarchical order.

Table 11: Constraints in the "Equity & Inclusion" category

Code	Description
E1 Handicap	
E1-1	Authorised students with disabilities shall receive extra time in accordance with accessibility policy.
E1-2	Students with disabilities shall be scheduled in rooms that meet accessibility standards (ramps, adapted desks, etc.).
E1-3	Required assistive equipment (e.g., screen readers, enlarged-font papers) shall be available at the examination venue.
E1-4	Dedicated support staff shall be assigned to assist candidates with disabilities during examinations.
E1-5	Accessible examination rooms shall be reserved for invigilators with disabilities.
E1-6	Students with disabilities shall be scheduled in separate rooms.
E2 Religion / Culture	
E2-1	Religious convictions must be respected.
E2-2	Where required, female proctors shall be assigned to examinations held for female-only cohorts. (Equivalent to I4-4)
E2-3	If an examination falls on a state-specific public holiday observed by any registered candidate, the examination shall be moved to an alternative slot. (Equivalent to L1-7)
E3 Limited Time	
E3-1	The scheduling algorithm shall respect the declared availability windows of students on work–study (alternance) programmes.
E3-2	The availability constraints of part-time students shall be honoured when allocating examination periods.
E3-3	Designated student groups may be restricted to a predefined subset of examination periods.
E4 Language	
E4-1	Foreign-exchange students shall, where feasible, be seated in separate rooms from domestic cohorts.
E4-2	The assigned invigilator shall be proficient in the language of the examination.
E4-3	Where practicable, examinations delivered in different languages should not occur on the same day for the same cohort of students.
E5 Diversity	
E5-1	Certain examination rules may apply only to specified categories of classes or students (e.g., clinical placements).
E5-2	Re-sit candidates should be scheduled in locations distinct from first-sit candidates.
E5-3	Students enrolled in adjacent-year curricula (AJAC) shall be treated as a distinct cohort for clash-avoidance purposes.

Table 12: Constraints in the "Logistics & Cost" category

Code	Description
L1 Window & Session	
L1-1	All examinations shall be scheduled within the official examination window published by the university.
L1-2	Designated examinations may be timetabled only inside a predefined subset of periods.
L1-3	The length of the overall examination session shall be minimised with respect to the number of calendar days required.
L1-4	The total number of examination days shall be minimised subject to feasibility.
L1-5	Each examination period shall have a minimum duration prescribed by institutional policy.
L1-6	Every examination day shall consist of a set of pre-defined time periods.
L1-7	If an examination falls on a state-specific public holiday observed by any registered candidate, the examination shall be moved to an alternative slot. (Equivalent to E2-3)
L1-8	Any free examination periods shall be placed at the end of the timetable.
L1-9	Maximise the number of exams that could be moved to another period without disrupting the rest of the schedule.
L2 Timing & Buffers	
L2-1	Timetables shall allow sufficient travel time for students moving between distinct campuses or sites.
L2-2	An examination shall not be split across multiple periods.
L2-3	The duration of an examination assigned to a period shall not exceed the duration of that period.
L2-4	A minimum transfer buffer shall separate two examinations when a candidate moves between examination centres. (Equivalent to L5-5)
L3 Location	
L3-1	Students should, where practicable, be examined on their home site.
L3-2	Consecutive examinations for the same student should not be held at different sites.
L3-3	Each candidate shall be assigned to a specific examination centre. (Equivalent to L5-4)
L3-4	Scheduling shall account for parking capacity and expected traffic conditions around examination venues.
L3-5	Each student shall be assigned to the geographically nearest examination centre, minimising travel distance.
L3-6	Proctors shall not be assigned to rooms they have flagged as unacceptable or unadapted.
L4 Cost	
L4-1	Usage of premium-cost resources (e.g., rented halls) shall be penalised in the objective function.
L4-2	Rooms with higher rental rates shall incur a cost penalty proportional to their use.
L4-3	Hiring temporary invigilation staff shall be modelled with a unit cost to be minimised.
L5 Pandemic	
L5-1	To mitigate pandemic risks, examinations should be assigned to department-specific time slots, limiting inter-department contact.
L5-2	High-capacity examinations should avoid early-morning slots to reduce congestion in public transport and lower transmission risk.
L5-3	Candidates may be separated in different centres to limit pandemic risk.
L5-4	Each candidate shall be assigned to a specific examination centre. (Equivalent to L3-3)
L5-5	A minimum transfer buffer shall separate two examinations when a candidate moves between centres. (Equivalent to L2-4)
L6 Repetition	
L6-1	Every examination must be scheduled.
L6-2	Every examination must be scheduled at least once.
L6-3	Every examination must be scheduled only once.

6 Conclusion

This study presents a comprehensive sourced catalogue of 167 exam timetabling constraints, derived from fieldwork, academic literature, and systematic construction. While many constraints are widely shared across institutions, others are shaped by local pedagogical, legal, or resource-specific factors. The hierarchical structure of the catalogue by domain and subcategory offers a modular framework that facilitates adoption, adaptation, and comparison across settings.

While the catalogue spans a broader mix of governance models and institution sizes, its geographic coverage is still incomplete. Europe, North America and the Middle East/Gulf dominate the sample, and only single institutions represent South America (Chile) and Africa (Tunisia); Oceania remains entirely absent. Major higher-education systems, such as those of Japan, Brazil, India, Pakistan, Mexico, Germany and Russia are likewise unrepresented. Capturing constraints from these regions would greatly strengthen the catalogue's global relevance.

To increase the practical value of this work, future directions could include identifying how best to select or prioritize constraints based on institutional context, one envisaged method is the wisdom of the crowd.

The index can already be used as a third-party to reduce mismatch of identification across teams and project but can also be the first steps towards a more rigid standard for industry and academia.

Declarations

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Competing interests The author declares that he has no relevant financial or non-financial interests to disclose.
Consent to participate Participants were informed of the purpose of the discussions and participated voluntarily.
Data availability The constraint catalogue and the origin table supporting the findings of this study are available at: https://github.com/Kaputchino/Constraints-for-the-Examination-Timetabling-Problem. The initial internal constraint list is available at: https://andreasmulard.fr/recherche/catalogue_contrainte.pdf.

Appendix A Journals

Publisher	Journal
Elsevier	Applied Soft Computing
	European Journal of Operational Research
	Technological Forecasting and Social Change
	Operations Research Letters
	Surveys in Operations Research and Management Science
	Operations Research Perspectives
INFORMS	Mathematics of Operations Research
	Information Systems Research
	Organization Science
	INFORMS Journal on Computing
	Service Science
	Operations Research
	Management Science
	Manufacturing and Service Operations Management
Springer	Central European Journal of Operations Research
	Operations Research / Computer Science Interfaces Series
	Operations Management Research
	Mathematical Methods of Operations Research
	International Series in Operations Research and Management Science
	Operations Research Forum
	Journal of the Operations Research Society of China
	Annals of Operations Research
	4OR
	Journal of Mathematical Modelling and Algorithms in Operations Research
	Operational Research
Taylor & Francis	Journal of the Operational Research Society
	Journal of Information Technology Case and Application Research
Wiley	International Transactions in Operational Research
	Advances in Operations Research
EDP Sciences	RAIRO – Operations Research
Operations Research Society of Japan	Journal of the Operations Research Society of Japan
Croatian Operational Research Society	Croatian Operational Research Review
University of Belgrade	Yugoslav Journal of Operations Research
University of the Punjab	Pakistan Journal of Statistics and Operation Research
Wrocław University of Science and Technology	Operations Research and Decisions

Table A1 Non-exhaustive list of journals in related fields by publisher

Appendix B Origin

Table B2: Origin of constraints

Origin	Constraint codes
Ankara HBV University (Bayar and Uzun Bayar (2020))	S1-5, S2-3, S2-4, S2-5
Athens University of Economics and Business (Dimopoulou and Miliotis (2001))	L1-3, L1-6, L6-1
Budapest University of Technology and Economics (Erdős and Kovári (2021))	I2-5, I2-6, I2-7
Çalık et al. (2024)	L3-5
Carleton University (Awad and Chinneck (1998))	I1-11, I1-12, I1-13, I1-14, I1-15, I1-16, I3-8, I4-6, L3-6
Carter and Laporte (1996)	D1-10, D3-6, L1-9
Cedar Crest College (Mehta (1981))	D1-11
Constructed constraint	D1-2, D2-2, D2-3
Cornell University (Ye et al. (2025))	R1-3
Di Gaspero and Schaerf (2001)	S1-15
École de technologie supérieure (Wong et al. (2002))	D2-1
École des Mines de Nantes (David (1998))	E4-2, L1-8
Ghaffar et al. (2025)	I2-2, I2-3
Hubei University of Technology (OuYang and Chen (2010))	R1-6
Israel Institute of Technology (Strichman (2017))	R3-1
Kamisli Ozturk et al. (2024)	S1-8, S1-10, D3-2, L5-1, L5-2
Katholieke Hogeschool Sint-Lieven (Demeester et al. (2012))	R1-5, R4-4, D1-13
King Fahd University of Petroleum and Minerals (Almaslami et al. (2023))	D3-7
King Saud University (Al-Negheimish et al. (2018); Hosny (2012))	I1-1, I1-2, I1-3, I1-6
Komijan and Koupaei (2012)	S1-2
Kutztown University (Sienko et al. (2016))	D2-5
Kuwait University (Al-Yakoob et al. (2010))	I3-4, I4-4, E2-2, L3-4
MARA University of Technology (Cowling et al. (2002); Kendall and Hussin (2005))	S1-13, I1-4, I1-8, I1-9, I2-1, I3-2, I3-3, I4-5, E2-3, L1-7
Masaryk University (Kochaniková and Rudová (2013); Rudová et al. (2014))	S3-1, S3-2, S3-3, I1-10, I3-6
Masdar Institute (Siew et al. (2024))	L1-4
Middle East Technical University (Ergül (1996))	S1-11
Modirkhorasani and Hoseinpour (2024)	L2-4, L3-3, L5-3, L5-4, L5-5
Muklason et al. (2017)	S1-12
Swansea University (Thompson and Dowsland (1995))	R1-7
United States Military Academy in West Point (Wang et al. (2010))	S2-6, D1-4, D1-12, E5-1, L6-2
Universidad Diego Portales (Cataldo et al. (2017))	S1-16, S2-7, S2-8, I3-7
Université Catholique de l'Ouest (Boizumault et al. (1996))	S1-3, D1-1, D2-4
Université de Sfax (Dammak et al. (2017))	S1-4
Université de Technologie de Compiègne (Arbaoui et al. (2013))	R4-1, D1-3, L2-2, L2-3
Université de Tours	R1-1, R1-2, R1-4, R2-1, R2-2, R2-3, R2-4, R2-5, R3-2, R4-2, S1-1, S2-1, S2-2, I3-1, I3-5, I4-3, D1-6, D1-7, E1-1, E1-2, E1-3, E1-4, E1-5, E1-6, E5-3, L1-1, L1-5, L4-1, L4-2, L4-3
Universiti Kebangsaan Malaysia (Ayob et al. (2007))	R3-7, R4-3
Universiti Malaysia Pahang (Mohmad Kahar and Kendall (2014))	I1-5, I1-7
University College Cork (Genc and O'Sullivan (2020))	S1-6
University Malaysia Sarawak (Siew et al. (2025))	R3-9, R4-7
University of British Columbia (Hare and Hamilton (2022))	D1-14
University of Sharjah (Lulu et al. (2024))	R3-8

Continued on next page

Origin	Constraint codes
University of Toronto and Carleton University (Carter et al. (1994))	S1-14, D3-5, E3-3
UK institutions (Burke et al. (1996))	R3-3, R3-4, R3-5, R4-5, R4-6, R4-7, S1-9, D1-5, D1-8, D1-9, D3-1, D3-3, E2-1, E3-1, E3-2, E4-1, E4-3, E5-2, L1-2, L2-1, L3-1, L3-2
Vestfold County School Department (Avella et al. (2025))	I2-4, I4-7
Yıldız Technical University (Güler and Geçi˙Ci˙ (2020); Güler et al. (2021))	R3-6, S1-7

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